PHYS 6810 Applied Statistics and Data Analysis in Physics

Spring Semester, 2024

3 Credits

Course Time: Tuesday/Thursday 02:20PM - 03:35PM

Instructors: Profs. Alexander van der Horst, Michael Doering, Oleg Kargaltsev

Course Place: Corcoran Hall, room 404A

Office Hours: TBD

Office: Corcoran 409 (van der Horst), 307 (Doering), 407 (Kargaltsev)

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Online semester start: see page 4 for details

Prerequisites:

Multivariable Calculus (e.g., MATH 2233 or equivalent), Linear Algebra (e.g., MATH 2184 or equivalent); calculus-based introductory physics (e.g., PHYS 1021,1022, and 2023 or equivalent physics courses).

Recommended Background:

Programming experience and good working knowledge of *either* Matlab, Mathematica, Python, IDL, or R. *You will need to have a laptop for this class.*

Synopsis

After taking this course, students will be able to explain the challenges and the best practices in statistical inference methods applied to physical science data, apply modern statistical methods, and create informative and appealing visualizations of the data and inferred statistically sound trends, correlations and dependencies. Students will make extensive use of real data and numerical methods to illustrate abstract statistical concepts.

Students will complete projects involving hands-on analysis of various datasets as part of lab-like class activities (working in groups) and individual homework assignments. The overarching goal of this course is to help students to develop analytical and practical skills for physical (and other) data analysis and interpretation using solid statistical methods well recognized within the discipline.

Recommended (but not required) reading materials for this course:

Herman J.C. Berendsen, "A Student's Guide to Data and Error Analysis"

P. Bevington & D.K. Robinson, "Data Reduction and Error Analysis for the Physical Sciences"

G. Bohm & G. Zech, "Introduction to Statistics and Data Analysis for Physicists"

John E. Freund, "Mathematical Statistics with Applications"

Campesato, O., "Python 3 for Machine Learning"

Brownlee, J., "Probability for Machine Learning - Discover How To Harness Uncertainty With Python"

Ž. Ivezić, A.J. Connolly, J. VanderPlas & A. Gray, "Statistics, Data Mining, and Machine Learning in Astronomy: A Practical Python Guide for the Analysis of Survey Data"

Web Site Resources:

http://www-library.desy.de/preparch/books/vstatmp_engl.pdf

http://astrostatistics.psu.edu/

https://towardsdatascience.com/machine-learning/home

https://towardsdatascience.com/data-science/home

Mathematica license for GW students: https://it.gwu.edu/wolfram-mathematica or https://seascf.seas.gwu.edu/wolfram-mathematica

Mathematica tutorial: http://home.gwu.edu/~kargaltsev/teaching/Mathematica_tutorial.nb

Python tutorial: : http://home.gwu.edu/~kargaltsev/teaching/Python_tutorial.ipynb
Addional resources are on the course page in the Blackboard: https://blackboard.gwu.edu/

Table 1. Course Objectives

Content Objectives	Skills Objectives
1. Be able to explain and discuss: ▪ the need and the nature of statistical data analysis ▪ be able to identify, explain, and critically evaluate the statistical methods used in the research papers related to physical sciences 2. Demonstrate deep understanding of fundamental statistical principles applicable to data analysis in physics and related disciplines 3. Acquire knowledge of a broad spectrum of modern statistical software tools.	1. Given a physical problem identify the statistical methods required to evaluate the impact of the uncertainties on the measurement or the choice of model 2. Identify and use appropriate software tools to carry out statistical analysis of various problems 3. Be able to apply statistical methods to a broad range of problems including those outside Physics (e.g., finance, signal processing)
Assessments and tools: ▪ Graded homework assignments ▪ Graded in-class projects assigned to groups ▪ Graded Final Project	

Course Schedule. A preliminary schedule of topics for the course is shown in the table below. Multiple instructors involved in the course will enrich the course from their

research in different areas of physics and astronomy, and graduate students and post docs already involved in research may be invited to classes to give guest lectrues.

Table 2. Classroom Schedule (approximate) and Activities

Class dates	Topics for Class Lectures and Activities
Week 1 (Jan 16 & 18)	Uncertainties and errors in physical data. Real life examples. Laws of Probability. Software tools
Week 2 (Jan 23 & 25)	Random variables, Expectation value, (co)variance, Practical applications.
Week 3 (Jan 30 & Feb 1)	Distributions and samples.
Week 4 (Feb 6 & 8)	Binomial, Poisson, and Normal Distributions. Central Limit Theorem. Student's t-distribution, F-distribution and F-test.
Week 5 (Feb 13 & 15)	The chi-squared and its minimization, chi-squared test. Maximum Likelihood.
Week 6 (Feb 20 & 22)	Best-fit parameter uncertainties, confidence intervals, covariance, and multidimensional fitting. Non-parametric statistics. Practical examples of fitting with real data.
Week 7 (Feb 27 & 29)	Bootstrap and Monte Carlo methods. Practical applications.
Week 8 (Mar 5 & 7)	Bayes' theorem. Bayesian Information Criterion.
Week 9 (Mar 19 & 21)	Bayesian Statistics. Least Absolute Shrinkage and Selection Operator.
Week 10 (Mar 26 & 28)	Time series analysis. Fourier, Wavelets, and other methods. Practical applications.
Week 11 (Apr 2 & 4)	Clustering and classification. Spatial statistics.
Week 12 (Apr 9 & 11)	Supervised & unsupervised machine learning with applications to classification. Generative AI and its applicaitons.
Week 13 (Apr 16 & 18)	Practical applications of clustering and classification.
Week 14 (Apr 23 & 25)	Data visualization techniques, linked view, dynamic updating. Practical applications.
Final: Apr 29	Final project presentations.

Course Format

Students are expected to attend classes in person. Requests for remote attendance can be made to the instructors on a case-by-case basis. This course will be a mix of lectures and lab-like activities. The activities can involve collaborative work in groups. There will be a focus on applying the theoretical methods and concepts described in the lectures, both in class and in homework assignments. The class time is the best opportunity for asking questions about difficult concepts. You will be strongly encouraged to ask questions and initiate discussions in class.

Evaluation Criteria (how the course grade will be determined)

The total grade for the course will be calculated with the following breakdown:

In-class Activities + Homework Assignments - Doering: 20%

In-class Activities + Homework Assignments - van der Horst: 20%

In-class Activities + Homework Assignments - Kargaltsev: 20%

Final Project: 30%

Participation: 10%

In class activities (vary by instructor): A mix of interactive discussions, quizzes, and short projects with your preferred software (choices are: Matlab, Mathematica, Python, IDL, or R).

Individual Homework: Regular homework assignments will include short computational projects to ensure good understanding of statistical methods and develop practical application skills.

Final Project: A take-home project that will involve solving a data-based problem using statistical and computational methods.

NOTE: *Use of generative AI (such as ChatGPT or other similar tools) is allowed to the extent specified by your course instructors. If no AI-specific instructions are given, generative AI should not be used for any tasks in this course. Any unauthorized use of generative AI will be a violation of Academic Integrity Code.*

Numerical course grades translate into letter grades using the following scale:

≥ 94.00	**A**	74.00-77.99	**C+**
90.00 – 93.99	**A-**	70.00 – 73.99	**C**
86.00 – 89.99	**B+**	66.00 – 69.99	**C-**
82.00 – 85.99	**B**	≤ 65.99	**F**
78.00 – 81.99	**B-**		

Distribution of hours spent on coursework

There will be two 75-minute lectures per week, totaling 150 minutes per week. Out-of-class coursework is estimated to be 5 hours per week, which consists mainly of doing homework and working on the final project.

Absences and Excuses

All requests to have an absence excused must follow standard University policy. If you will be missing a class you must infrom your instructor ahead of the class and provide a legitimate reason for your absence.

Academic Integrity

Academic integrity is an essential part of the educational process, and all members of the GW community take these matters very seriously. As the instructors of record for this course, our role is to provide clear expectations and uphold them in all assessments. Violations of academic integrity occur when students fail to cite research sources properly, engage in unauthorized collaboration, falsify data, and otherwise violate the Code of Academic Integrity. If you have any questions about whether particular academic practices or resources are permitted, you should ask us for clarification. If you are reported for an academic integrity violation, you should contact the Office of Student Rights and Responsibilities (SRR) to learn more about your rights and options in the process. Consequences can range from failure of assignment to expulsion from the university and may include a transcript notation. For more information, please refer to the SRR website (https://studentconduct.gwu.edu/academic-integrity), email rights@gwu.edu, or call 202-994-6757.

Religious Holidays

Students must notify faculty during the first week of the semester in which they are enrolled in the course, or as early as possible, but no later than three weeks prior to the absence, of their intention to be absent from class on their day(s) of religious observance. If the holiday falls within the first three weeks of class, the student must inform faculty in the first week of the semester. For details and policy, see "Religious Holidays" at provost.gwu.edu/policies-procedures-and-guidelines.

Electronic Course Materials & Class Recordings

Students are encouraged to use electronic course materials, including recorded class sessions, for private personal use in connection with their academic program of study. Electronic course materials and recorded class sessions should not be shared or used for non-course related purposes unless express permission has been granted by the instructor. Students who impermissibly share any electronic course materials are subject to discipline under the Student Code of Conduct. Please contact the instructors if you have questions regarding what constitutes permissible or impermissible use of electronic course materials and/or recorded class sessions. Please contact Disability Support Services at disabilitysupport.gwu.edu if you have questions or need assistance in accessing electronic course materials.

Support for Students Outside the Classroom

Disability Support Services (DSS) 202-994-8250

Any student who may need an accommodation based on the potential impact of a disability should contact Disability Support Services at disabilitysupport.gwu.edu to establish eligibility and to coordinate reasonable accommodations.

Counseling and Psychological Services 202-994-5300
GW's Colonial Health Center offers counseling and psychological services, supporting mental health and personal development by collaborating directly with students to overcome challenges and difficulties that may interfere with academic, emotional, and personal success. healthcenter.gwu.edu/counseling-and-psychological-services.

Security

- In an emergency: call GWPD 202-994-6111 or 911
- For situation-specific actions: https://safety.gwu.edu/emergency-response-handbook
- In an active violence situation: Get Out, Hide Out, or Take Out; see https://go.gwu.edu/shooterpret
- Stay informed: https://safety.gwu.edu/stay-informed